

Comparison of EKF and UKF State Estimation with Measurement Gating and RK4 Propagation

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Abstract—This report evaluates an extended Kalman filter (EKF) and an unscented Kalman filter (UKF) for the nonlinear vehicle-state estimation problem posed in the EMCH792 final project. The analysis used the provided deployment dataset, estimated sensor variances directly from the calibration sets, and compared rectangular propagation, scalar measurement gating, and fourth-order Runge-Kutta (RK4) propagation. The calibration pass yielded $R_1 = 0.087247$ and $R_2 = 0.006536$ with a constant sample period of 0.100 s. A short sweep over initial covariance scale and scalar chi-square gate threshold selected a covariance scale of 0.50 and a 99% gate at 6.6349. Across the full comparison, the rectangular UKF outperformed the rectangular EKF in ungated accuracy, while the RK4 UKF achieved the best overall position error at 1.4206 m. The gating logic removed the largest y_2 outliers and improved heading estimation, but excessive late rejection of y_1 measurements increased position error in the gated runs.

Index Terms—Extended Kalman filter, Unscented Kalman filter, Nonlinear state estimation, Measurement gating, Runge-Kutta integration

I. PROBLEM I. CALIBRATION VARIANCE ESTIMATION

The first task was to estimate the two sensor variances from the calibration data and lock the deployment sample period. The calibration pass produced $R_1 = 0.087247$ for the range-like measurement and $R_2 = 0.006536$ for the cubic heading sensor. The deployment dataset remained uniformly sampled at 0.100 s, which made the rectangular covariance propagation required by the assignment directly applicable. The filters were initialized from the first truth sample so that the comparison emphasized filter behavior, rather than startup mismatch, across the 100 available samples.

II. PROBLEM II. EKF AND UKF FORMULATION

Both filters used the assignment state $x = [x, y, \psi, u]^T$ with steering angle and longitudinal acceleration as inputs, fixed geometry $l_f = l_r = 1$, and the prescribed continuous-time process covariance. The EKF used the rectangular Euler prediction and first-order Jacobian linearization for both process and measurement updates. The UKF used the scaled unscented transform with $\alpha = 10^{-3}$, $\beta = 2$, and $\kappa = 0$, propagated each sigma point through the selected integrator, and reconstructed mean and covariance from the propagated sigma set. In every configuration, the two measurements were

processed sequentially as scalar updates so that each sensor could be accepted or rejected independently.

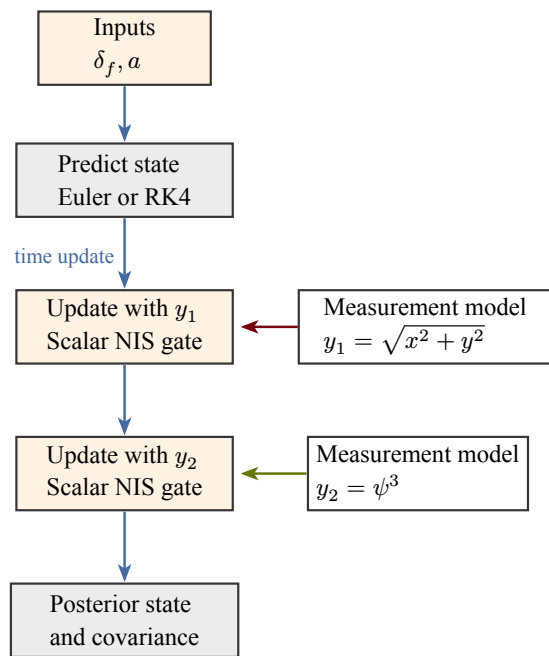


Fig. 1. Standalone CeTZ diagram of the sequential estimation workflow. The figure was authored separately, compiled to PDF, and then imported into the main IEEE-formatted document as a standard image asset.

The ungated rectangular comparison favored the UKF. The EKF rectangular run produced a position RMSE of 3.6891 m with a heading RMSE of 17.8381 deg, while the rectangular UKF reduced these values to 1.6879 m and 4.9532 deg. This result is consistent with the nonlinear measurement model, especially the ψ^3 sensor, where the UKF preserved nonlinear curvature more effectively than the first-order EKF approximation.

III. PROBLEM III. MEASUREMENT GATING STUDY

The tuning sweep compared two reasonable scalar chi-square thresholds, namely 95% and 99%, across five initial covariance scales. The selected setting was the 99% gate with a 0.50 scaling on the base initial covariance because

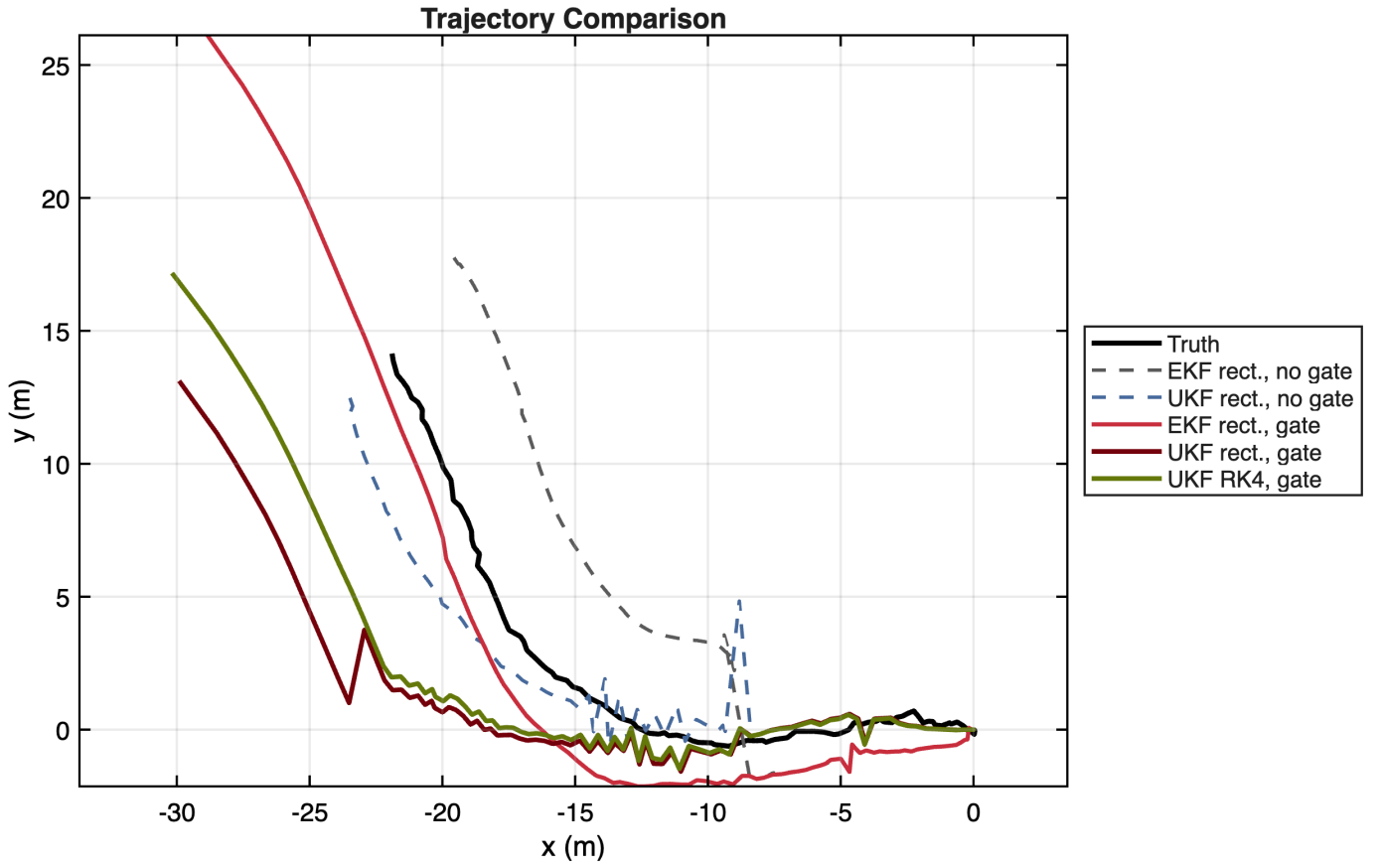


Fig. 2. Truth and estimated trajectories for the main filter configurations. The ungated UKF remains closest to the measured path, while the gated runs become increasingly prediction dominated late in the record.

it minimized the composite score across the gated EKF and UKF rectangular runs.

The gate did exactly what it was intended to do for the heading-like sensor. The faulty y_2 measurement was rejected three times in each final gated run, which reduced the heading RMSE from 17.8381 deg to 5.0388 deg for the EKF and from 4.9532 deg to 3.8400 deg for the UKF. However, the

same logic also rejected many late y_1 measurements, namely 24 in the EKF and 14 in the UKF. As a consequence, the position RMSE increased to 3.8951 m and 4.0885 m for the gated rectangular filters. The practical interpretation is that the scalar normalized innovation squared test was effective at flagging the strongest outliers, but the process model was not accurate enough to fully replace the range-like measurement

TABLE I
INITIALIZATION-SCALE AND GATE-THRESHOLD SWEEP USED TO SELECT THE FINAL GATED COMPARISON SETTINGS.

P0 scale	Gate	Threshold	EKF pos.	UKF pos.	EKF psi deg	UKF psi deg	Score
0.50	95%	3.8415	13.6143	3.7374	8.3801	3.5998	2.4823
0.50	99%	6.6349	3.8951	4.0885	5.0388	3.8400	1.8387
1.00	95%	3.8415	16.8178	4.8416	6.1110	3.7884	2.8714
1.00	99%	6.6349	16.8178	4.5638	6.1110	3.7879	2.8548
2.00	95%	3.8415	16.6026	8.5215	4.0514	4.6785	3.0537
2.00	99%	6.6349	15.6909	6.3686	3.9223	4.7091	2.7935
4.00	95%	3.8415	16.5372	8.3965	3.4015	5.0742	3.0016
4.00	99%	6.6349	15.7881	6.5075	3.6325	5.0731	2.7646
8.00	95%	3.8415	17.3709	9.4532	4.9371	4.4273	3.0173
8.00	99%	6.6349	15.7888	5.4135	4.8713	4.4215	2.4947

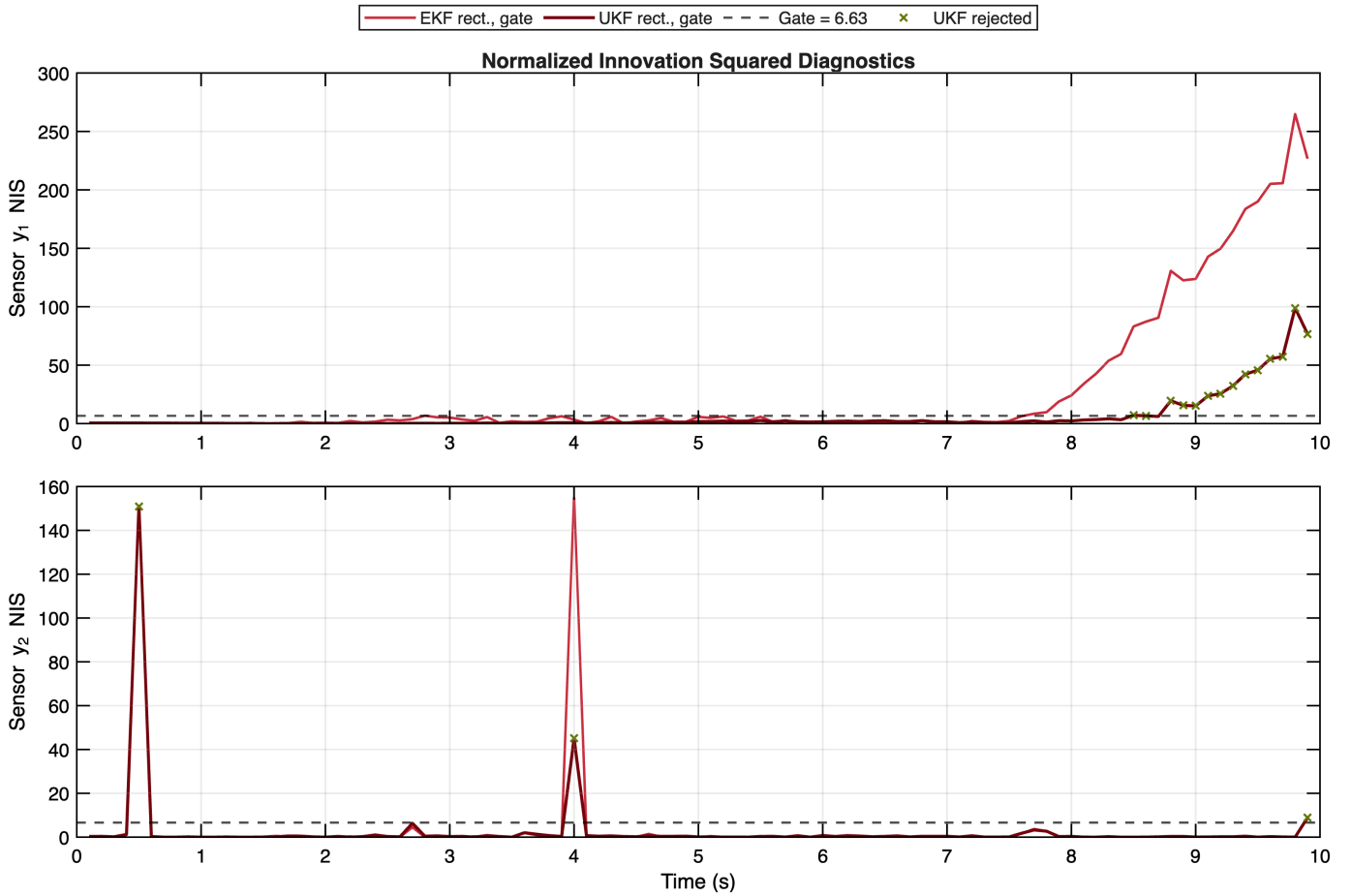


Fig. 3. Normalized innovation squared histories for both scalar sensors. The isolated y_2 spikes are clearly detectable, while the late growth in y_1 residuals explains the large number of range-measurement rejections.

once repeated rejections accumulated near the end of the maneuver.

IV. PROBLEM IV. RK4 PROPAGATION FOR THE UKF

Replacing the UKF rectangular propagation with RK4 improved the ungated accuracy. The ungated RK4 UKF achieved the best position RMSE in the study at 1.4206 m, improving on the ungated rectangular UKF by 0.5680 m. Under the same gating logic, the RK4 UKF also delivered the best heading RMSE among the gated UKF runs at 3.7895 deg, although the persistent late measurement rejections still limited the position benefit in the gated case.

The covariance histories support that interpretation. The UKF carried a larger uncertainty envelope than the EKF, which is expected for a sigma-point method operating on the same nonlinear dynamics and measurement set. The RK4 propagation also produced a slightly smoother covariance evolution than the rectangular UKF during the final part of the trajectory.

V. PROBLEM V. RUNTIME COMPARISON

The runtime study used sixty repeated executions per configuration so that transient timing noise would not dominate the comparison. The EKF remained the cheapest method at

2.097 ms per run on average, the rectangular UKF required 3.879 ms, and the RK4 UKF required 4.641 ms. The RK4 propagation therefore added about 0.763 ms relative to the rectangular UKF. That cost increase is modest compared with the gain in ungated trajectory accuracy.

TABLE II
AVERAGE RUNTIME OVER SIXTY REPEATED EXECUTIONS FOR THE PRINCIPAL ESTIMATOR CONFIGURATIONS.

Configuration	Mean ms	Std ms
EKF rect., gate	2.097	0.958
UKF rect., gate	3.879	0.157
UKF RK4, gate	4.641	0.022

VI. PROBLEM VI. OVERALL COMPARISON AND CONCLUSIONS

The overall comparison is clear. The UKF was the stronger estimator for this dataset because the nonlinear process and measurement relationships were significant enough that the EKF linearization lost accuracy. The covariance comparison also favored the UKF qualitatively because the sigma-point filters carried larger and more credible uncertainty levels in-

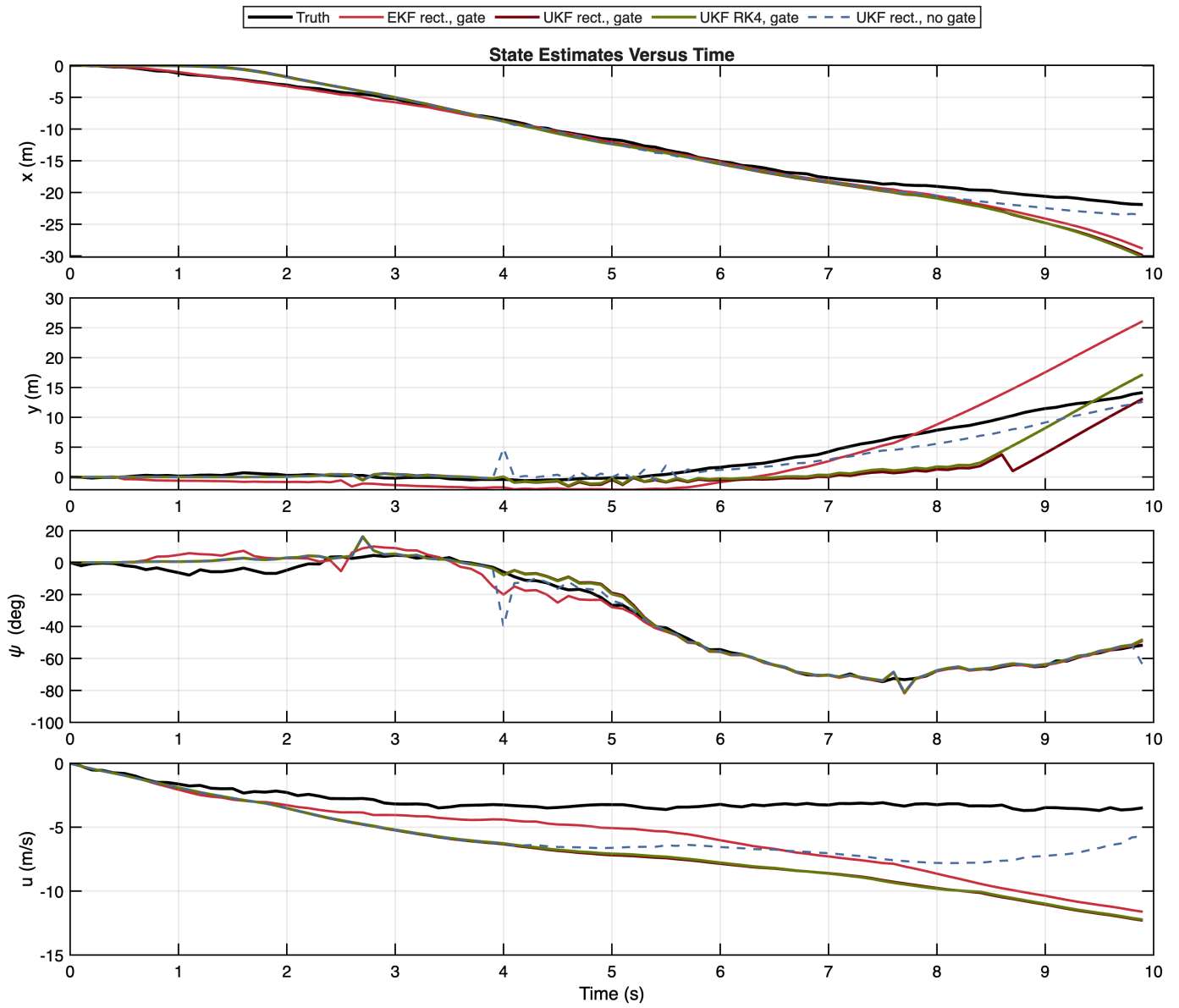


Fig. 4. State estimates versus time for the principal comparison cases. The gated filters maintain improved heading agreement, but their position and speed estimates drift as measurements are rejected.

stead of the tighter EKF covariance trace. Measurement gating helped where it mattered most by removing the strongest y_2 outliers and reducing heading error, but it also exposed the weakness of the process model by forcing the filters to coast through too many late y_1 rejections. Finally, the RK4 propagation was the cleanest algorithmic improvement in the study because it produced the best overall ungated trajectory while increasing runtime only slightly.

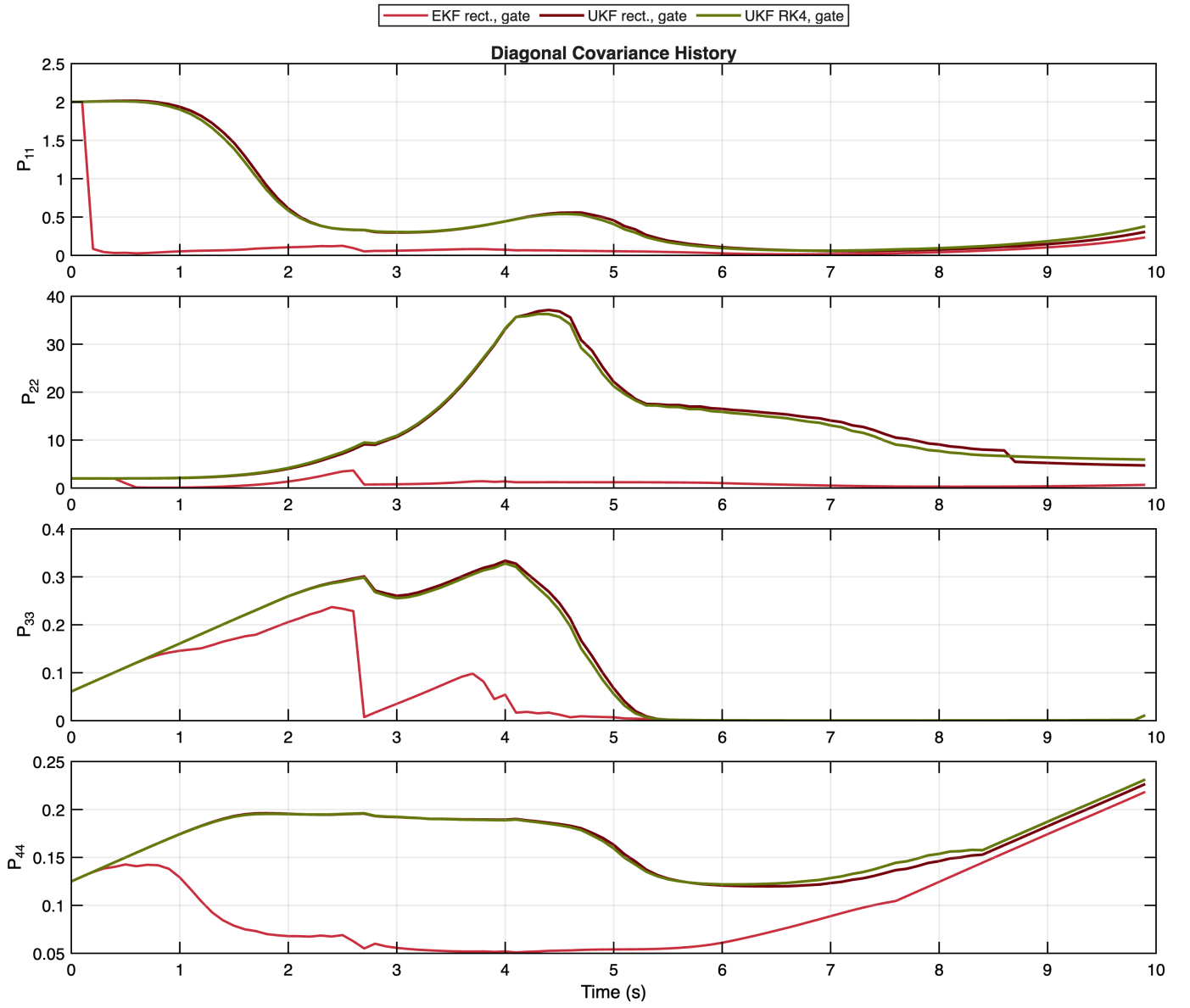


Fig. 5. Diagonal covariance histories for the gated EKF, gated rectangular UKF, and gated RK4 UKF. The UKF variants retain visibly larger uncertainty than the EKF, which is consistent with their stronger nonlinear treatment.

TABLE III
STATE-ESTIMATION ACCURACY, COVARIANCE SUMMARY, AND MEASUREMENT-REJECTION COUNTS FOR ALL EVALUATED CONFIGURATIONS.

Configuration	x	y	psi deg	u	pos.	avg tr(P)	rej y1	rej y2
EKF rect., no gate	1.1828	3.4943	17.8381	1.5688	3.6891	0.7417	0	0
UKF rect., no gate	1.0710	1.3046	4.9532	2.9613	1.6879	6.0709	0	0
EKF rect., gate	1.8305	3.4382	5.0388	3.6671	3.8951	1.1537	24	3
UKF rect., gate	2.2324	3.4252	3.8400	4.5974	4.0885	13.0001	14	3
UKF RK4, no gate	0.9634	1.0441	5.0272	2.8578	1.4206	5.9494	0	0
UKF RK4, gate	2.2670	2.6936	3.7895	4.5681	3.5206	12.7821	15	3